HAna Note

1 Sup Inf & Set || Nested Intervals || Sequence & Bolzano Weierstrass

1.1 Supremum and Infimum & Set

Completeness Axiom: If $\exists$ UP $\Rightarrow$ LUB (sup) ; If $\exists$ LB $\Rightarrow$ GLB (inf). **Archimedean Property**: $\forall x \in \mathbb{R}^+ \Rightarrow \exists n \in \mathbb{N}\ s.t.\ n > x$.

Countable Set: A set E is countable if $\exists$ bijection $f : \mathbb{N} \to E$.

· **Property**: Let $(E_n)_{n\in\mathbb{N}}$ be countable sets. $\Rightarrow$ $^1 \bigcup E_n$ is countable. (Can infinite) $^2 \prod E_n$ is countable. (Finite) $^3[0,1]$ uncountable.

$^\oplus$ **Dense Set**: Set A is dense in $\mathbb{R}$ iff $\forall x \in \mathbb{R}, \forall \varepsilon > 0\ \exists a \in A$ s.t. $|x-a| < \varepsilon$ e.g.$^\oplus$ $\mathbb{Q}$ is dense in $\mathbb{R}$

$^\oplus$ **Power Set** $\mathcal{P}$: $P(E)$ is the set of all subsets of E. ps: $\mathcal{P}(\mathbb{R})$ is the set of all subsets of $\mathbb{R}$.

1.2 Nested Intervals

Def of $\lambda(I)$: Let I be an interval. $\lambda(I) := \sup I - \inf I$ || $\lambda((a,b)) = \lambda([a,b)) = \lambda((a,b]) = \lambda([a,b]) = b-a$.

Nested Intervals: Sequence $(I_n)_{n\in\mathbb{N}}$ is nested if $I_n \supseteq I_{n+1}$ ($\downarrow$) for all $n \in \mathbb{N}$.

Nested Interval Theorem: If $(I_n)_{n\in\mathbb{N}}$ is nested *closed bounded* interval sequence $\Rightarrow E := \bigcap I_n \neq \emptyset$.

· **Proeprty**: ^{1}If $\lim \lambda(I_n) = 0 \Rightarrow E = a$ $^{2\oplus}$If $I_n = [a_n, b_n]$ $\Rightarrow$ $E = [\sup a_n, \inf b_n]$

1.3 Sequence, Cauchy & Bolzano Weierstrass Theorem

Useful Seq Limit: $\lim \frac{1}{n^p} = 0, p > 1$; $\lim a^n = 0, |a| < 1$; $\lim n^{\frac{1}{n}} = 1$; $\lim a^{\frac{1}{n}} = 1, a > 0$

Operation in Seq Limit: *Squeeze Theorem* ; *Monotone Convergence Theorem (MCT)* ; $\lim \alpha x_n = \alpha \lim x_n$

· $\lim(x_n + y_n) = \lim x_n + \lim y_n$; $\lim x_n \cdot y_n = \lim x_n \cdot \lim y_n$; $\lim \frac{x_n}{y_m} = \frac{\lim x_n}{\lim y_n}, y_n \neq 0$

Comparison Theorem (For Seq): If $\lim x_n, y_n$ exist, and $x_n \leq y_n$ for all $n \geq N$ then $\lim x_n \leq \lim y_n$.

Cauchy Sequence: Sequence $(a_n)_{n\in\mathbb{N}}$ is Cauchy iff $\forall \varepsilon > 0, \exists N \in \mathbb{N}\ s.t.\ |a_n - a_m| < \varepsilon$ for all $m, n \geq N$. || **Convergent $\Leftrightarrow$ Cauchy**

Bolzano-Weierstrass Theorem: Every *bounded sequence* has a *convergent subsequence*.

Def of limit superior and inferior: $\limsup a_n := \lim(sup_{k\geq n} a_k)$ || $\liminf a_n := \lim(inf_{k\geq n} a_k)$

· **Property**: If $(a_n)_{n\in\mathbb{N}}$ bounded. $\Rightarrow$ $^{\oplus 1}\exists(a_{n_k}) \to \limsup a_n$ Hint:$\exists a_{n_k}, \sup_{j\geq k} a_j - \frac{1}{k} \leq a_{n_k} \leq \sup_{j\geq k} a_j, \sup_{j\geq n_k} a_j \leq \sup_{j\geq k} a_j$ $^{\oplus 2}\exists(a_{n_k}) \to \liminf a_n$
$^{\oplus 3}\forall a_{n_k} \to a\ \liminf a_n \leq a \leq \limsup a_n$

Subsequence Theorem: In each case, the subsequence can be taken to be *monotonic*.

1. $\exists a \in \mathbb{R}\ s.t.\ \forall \varepsilon > 0$, there are infinitely many $n \in \mathbb{N}$ $s.t.\ |x_n - a| < \varepsilon$ (∃ 聚点 a) $\Leftrightarrow$ $\exists$ subsequence of (x_n) $s.t.$ $\lim\ x_{n_k} = a$
2. $(x_n)_{n\in\mathbb{N}}$ is not bounded above (below) $\Leftrightarrow$ $\exists$ subsequence of x_n $s.t.$ $\lim x_{n_k} = \infty\ (-\infty)$
3. If $\lim x_n = L \Rightarrow \forall x_{n_k}, \lim x_{n_k} = L$

2 Series

Cauchy Criterion for Series: $\sum_{k=1}^\infty a_k$ converges $\Leftrightarrow$ $\forall \varepsilon > 0, \exists N \in \mathbb{N}\ s.t.\ \forall n \geq m \geq N\ |\sum_{k=m+1}^n a_k| = |a_{m+1} + \cdots + a_n| < \varepsilon$.

p-Test: $\sum \frac{1}{n^p}$ convergent iff $p > 1$ **Root Test** **Ratio Test**$_{|n|^{1/n}, |n+1/n|}$ **Geometric Series**: $\sum p^n$ converges iff $|p| < 1$

Divergence Test: If $\lim a_n \neq 0 \Rightarrow \sum a_n$ diverges. (If $\sum a_n$ convergent $\Rightarrow \lim a_n = 0$.)

Comparison Test: If $0 < a_n < b_n, \sum b_n$ convergent $\Rightarrow \sum a_n$ also ; $\sum a_n$ divergent $\Rightarrow \sum b_n$ also.

· **Limit Comparison Test**: If $a_n, b_n \geq 0, \lim \frac{a_n}{b_n} = L > 0$ $\Rightarrow$ If $L \in (0, \infty)$: $\sum a_n$ convergent iff $\sum b_n$ also.
If $L = 0 : \sum b_n$ convergent $\Rightarrow \sum a_n$ also. $L = \infty : \sum b_n$ divergent $\Rightarrow \sum a_n$ also.

Integral Test: Let $f : [1, \infty) \to \mathbb{R}$ is 非负递减, $a_n = f(n)$. Then $\sum a_n$ converges iff $\int_1^\infty f(x)dx < \infty$.

Absolutely Convergence: $\sum a_n$ convergent absolutely iff $\sum |a_n|$ convergent. **If convergent abs $\Rightarrow$ convergent.**

Alternating Series Test: If a_n decreasing, $a_n \geq 0, \lim a_n = 0$. Then $\sum (-1)^{n-1} a_n$ convergent.

Cauchy's Condensation Test: If $a_n \geq 0, a_n$ decreasing, $\Rightarrow$ $[\sum a_n convergent \Leftrightarrow \sum 2^n a_{2^n}\ also\ ]$

Riemann Rearrangement Theorem: If $\sum_{k=1}^\infty a_k$ is conditionally convergent. $\Rightarrow$ $\forall s \in \mathbb{R}, \exists$ bijection $\sigma : \mathbb{N} \to \mathbb{N}$ s.t. $\sum_{n=1}^\infty a_{\sigma(n)} = s$.

Rearrangement for Abs Series: If $\sum_{n=1}^\infty a_n$ is absolutely convergent. $\Rightarrow$ $\forall$ bijection $\sigma : \mathbb{N} \to \mathbb{N}$, $\sum_{n=1}^\infty a_{\sigma(n)} = \sum_{n=1}^\infty a_n$.

3 Continuous & Differentiable Functions Properties

Def of lim of func: **1**.$\forall \varepsilon > 0, \exists \delta > 0\ s.t. 0 < |x - x_0| < \delta, \Rightarrow |f(x) - L| < \varepsilon$. **2**.$\forall x_n \neq x_0\ s.t. \lim(x_n) = x_0, then\ \lim f(x_n) = L$.

Def of cont of func: **1**.$\forall \varepsilon > 0, \exists \delta > 0\ s.t. \forall x, |x - x_0| < \delta \Rightarrow |f(x) - f(x_0)| < \varepsilon$. **2**.$\forall (x_n)_{n\in\mathbb{N}}, if\ \lim x_n = x_0 \Rightarrow \lim f(x_n) = f(x_0)$

Def of diffi: **1**.$f'(a) = \lim_{x\to a} \frac{f(x)-f(a)}{x-a}$ **2**.$f'(a) = \lim_{h\to 0} \frac{f(a+h)-f(a)}{h}$.

The Extreme Value Theorem (EVT): If $f : [a,b] \to \mathbb{R}$ is continuous, then f has a max and min value.

Intermediate Value Theorem (IVT): $f : [a,b] \to \mathbb{R}$ is continuo,$f(a) \neq f(b), y \in [f(a), f(b)]$.$\Rightarrow \exists c \in (a,b)\ s.t. f(c) = y$

IVT (for deriv) / (Darboux Theorem): If f differentiable in $I \Rightarrow \forall m \in (f'(a), f'(b))$ we have: $\exists c$ s.t. $f'(c) = m$

Rolle's Theorem: If f continuous on $[a,b]$, differentiable on (a,b) and $f(a) = f(b)$ Then: $\exists c \in (a,b)$ s.t. $f'(c) = 0$

Mean Value Theorem (MVT): Let f, g continuous on $[a,b]$ and differentiable on (a,b): ps: a=b 时依然成立

· **1**.$\exists c \in (a,b)$ s.t. $\frac{f(b)-f(a)}{b-a} = f'(c)$ **2**. $\exists c \in (a,b)$ s.t. $\frac{f(b)-f(a)}{g(b)-g(a)} = \frac{f'(c)}{g'(c)}$

Theorem: If f is 1-1 and continuous $\Rightarrow$ f is strictly monotone. and f^{-1} is continuous and strictly monotone.

Inverse Function Theorem: Let f differentiable in I (open interval), and $\forall x, f'(x) \neq 0$:

· **1.** $f(I)$ is open interval. **2.** f is invertible. **3.** f^{-1} is differentiable and $(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))}$

Second Deriv Test: f twice continuously differentiable in $I, f'(x_0) = 0$: **1.** $f''(x_0) > 0\ (< 0)$ local min (max) point.

4 Pointwise & Uniformly Convergence

4.1 Pointwise Convergence

Pointwise Convergence: Let $(f_n)_{n\in\mathbb{N}}$ be a sequence of functions. $f_n: E \to \mathbb{R}$ converge pointwise to $f: E \to \mathbb{R}$ if: 不加说明默认 pointwise

1. **Version 1**: $\forall x \in E, \forall \varepsilon > 0, \exists N_{\varepsilon,x}\ s.t. \forall n > N\ |f_n(x) - f(x)| < \varepsilon$
2. **Version 2**: $\forall x \in E, \lim f_n(x)$ exists and $\lim f_n(x) = f(x)$

Pointwise Convergence NOT Preserve: ^{1}*continuity*, ^{2}*differentiability*, $^3(\lim f_n)' \neq \lim f_n'$, $^4 \int \lim f_n \neq \lim \int f_n$

· 对应反例: $f_n: [0,1] \to \mathbb{R} := x^n$ $\quad ^2 f_n: (-1,1) \to \mathbb{R} := \sqrt{x^2 + \frac{1}{n}}$ $\quad ^3 f_n: (-1,1) \to \mathbb{R} := \frac{\sin(nx)}{n}$ $\quad ^4 f_n: (0,1] \to \mathbb{R} := \begin{cases} n, 0 < x \leq \frac{1}{n} \\ 0, \frac{1}{n} < x \leq 1 \end{cases}$

4.2 Uniformly Convergence

Graph: $\Gamma(f) := \{(x, f(x)) | x \in E\}$ ps: $f: E \to \mathbb{R}$

Vertical Neighbourhood $\mathcal{N}(f, \varepsilon) := \{(x,y) | x \in E, |y - f(x)| < \varepsilon\}$ ps: $f: E \to \mathbb{R}$

Uniformly Convergence: $f_n: E \to \mathbb{R}$ converge uniformly iff: || **Uniformly $\Rightarrow$ Pointwise**

1. **Version 1**: $\forall \varepsilon > 0, \exists N_\varepsilon\ s.t. \forall n > N, \forall x \in E, |f_n(x) - f(x)| < \varepsilon$
2. **Version 2**: $\lim \left[\sup_{x\in E} |f_n(x) - f(x)|\right] = 0$
3. **Version 3 | Uniformly Cauchy**: $\forall \varepsilon > 0, \exists N_\varepsilon \in \mathbb{N}\ s.t. \forall x \in E, m, n > N\ |f_n(x) - f_m(x)| < \varepsilon$ || **Converge Unif $\Leftrightarrow$ Unif Cauchy**
4. **Version 4 | Vertical Neighbourhood Test**: $\forall \varepsilon > 0, \exists N \in \mathbb{N}\ s.t. \forall n > N, \Gamma(f_n) \subset \mathcal{N}(f, \varepsilon)$

Sequence Test: If $f_n: E \to \mathbb{R}$ converge uniformly to f. $\Rightarrow$ If $(x_n)_{n\in\mathbb{N}} \in E, \lim |f_n(x_n) - f(x_n)| = 0$ 不能当作 $\forall x_n$

Properties of Uniformly Convergent: Let $f_n, f, g: I \to \mathbb{R}$ Then: NOT preserve: differentiable.

1. **Continuous**: If $^1 \lim f_n(x) = f(x)$ uniformly, $^2 \forall f_n$ continuous at x_0 $\Rightarrow$ f continuous at x_0. $\Downarrow \forall x \in I, (\lim f_n)'(x) = \lim f_n'(x)$
2. **Termwise differentiation**: If $^1 I$ is *bounded open*; $^2 \forall f_n$ diff, $\lim f_n' = g$ unif; $^3 \exists x_0 \in I\ s.t.\ f_n(x_0)$ converges. $\Rightarrow$ $\lim f_n = f$ unif; f' diff and $f' = g$.
3. Cannot use now If $^1 \lim f_n(x) = f(x)$ uniformly on *closed interval* $[a,b]$, $^2 f_n$ integrable on $[a,b]$ $\Rightarrow$ $\lim \int_a^b f_n(x)dx = \int_a^b \lim f_n(x)dx$
4. $^\oplus$ If $f_n: [a,b] \to \mathbb{R}, g_n: [a,b] \to \mathbb{R}$ converge uniformly. $\Rightarrow$ $\frac{f_n}{g_n}$ converge uniformly also. ps: (a,b) 不行 Hint: $f_n g - f g_n = f_n g - f g + f g - f g_n$

Def of Series of Functions: Let $f_n: E \to \mathbb{R}$; Def partial sum $s_n(x) := \sum_{k=1}^n f_k(x)$

1. $\sum f_n$ converge pointwise iff s_n converge pointwise. **2.** $\sum f_n$ converge uniformly iff s_n converge uniformly. **3.** $\sum f_n$ converge absolutely (pointwise) iff $\sum |f_n|$ converge (pointwise).

Weierstrass M-Test: $f_n: E \to \mathbb{R}, \exists (M_n)_{n\in\mathbb{N}} \geq 0$ s.t. $^1 \forall x \in E, \forall n \in \mathbb{N}: |f_n| < M_n$; $^2 \sum M_n$ converges. $\Rightarrow$ $\sum f_n$ converge absolutely & Uniformly.

5 Power Series

5.1 Power Series, Radius of Convergence, Continuity and Differentiability

Power Series: A series from $\sum_{n=0}^\infty a_n(x-c)^n$ **Center**: c. **Coefficients**: a_n.

Radius of Convergence (R): For a power series $\sum a_n(x-c)^n$, there are three situations for it's radius of convergence:

1. $R = 0$ if $\sum a_n(x-c)^n$ converges only $x = c$.
2. $R = \infty$ if $\sum a_n(x-c)^n$ converges absolutely $\forall x \in \mathbb{R}$.
3. $\exists R > 0$ s.t. $\sum a_n(x-c)^n$ converges absolutely if $|x-c| < R$; diverges if $|x-c| > R$.
4. **All-In-One**: $R = \sup\{r \geq 0 : (a_n r^n)_{n\in\mathbb{N}}\ is\ \underline{bounded}\}$

Properties | In Radius of Convergence: Assume $f(x) = \sum a_n(x-c)^n$ with $R > 0$.

1. **Continuous**: $f(x)$ continuous on $(c-R, c+R)$.
2. **Convergence**: $\forall 0 < r < R$, $f(x)$ converges absolutely & uniformly on $[c-r, c+r]$.
3. **Preserve R**: $\sum_{n=0}^\infty a_n(x-c)^n$ and $\sum_{n=1}^\infty n a_n (x-c)^{n-1}$ have same R. ps: 即求导 R 不变
4. **Termwise differentiation**: $f(x)$ is differentiable on $(c-R, c+R)$ || $^1 f'(x) = \sum_{n=1}^\infty n a_n (x-c)^{n-1}, x \in (c-R, c+R)$
 $^2 \forall 0 < r < R$, $f'(x)$ converges absolutely and uniformly on $[c-r, c+r]$. ps: 即可以对里面求导
5. **Coefficients & Diff**: $^1 f(x)$ is *infinitely differentiable* on $(c-R, c+R)$ || $^2 a_k = \frac{f^{(k)}(c)}{k!}$
6. $^\oplus$ **Uniqueness**: If $\sum a_n(x-c)^n = \sum b_n(x-c)^n$ for $\forall x \in \mathbb{R}, |x-c| < r_0$ $\Rightarrow$ $a_n = b_n$

5.2 Exponential Function & Real Analytic Function

Exponential Function: $E(x) := \sum_{n=0}^\infty \frac{x^n}{n!}$ with $R = \infty$ ps: usually denoted by $exp(x)$ or e^x

Real Analytic Function: $f: I \to \mathbb{R}$ is real analytic iff $\exists (a_n)_{n\in\mathbb{N}}$ s.t. $f(x) = \sum_{n=0}^\infty a_n(x-c)^n$ for $\forall x \in I$ i.e. Series converges pointwise on I.

Taylor Series: Given ^{1}*Open Interval* I, ^{2}*infinitely differentiable* $f: I \to \mathbb{R}$. $\Rightarrow$ Taylor Series of f at $c \in I := \sum_{n=0}^\infty \frac{f^{(n)}(c)}{n!}(x-c)^n$

· Not all infinitely differentiable functions are real analytic: e.g. $f(x) = \{e^{-1/x^2}, x > 0; 0, x < 0$

Taylor's Theorem: If $f: (a,b) \to \mathbb{R}$ is differentiable function at $x_0 \in (a,b)$:

1. $P_{n,f,x_0}(x) := f(x_0) + \sum_{k=1}^n \frac{f^{(k)}(x_0)}{k!}(x-x_0)^k$ (Taylor's polynomial of degree n at x_0)
2. If $f^{(n+1)}$ exists on (a,b), then: $\exists \xi \in (x, x_0)$ s.t. $f(x) = P_{n,f,x_0}(x) + \frac{f^{(n+1)}(\xi)}{(n+1)!}(x-x_0)^{n+1}$
3. Remainder $R_{n,f,x_0} := f(x) - P_{n,f,x_0}$. || $f: I \to \mathbb{R}$ is real analytic iff $\lim R_{n,f,x_0} = 0$ for $\forall x \in I$

6 Open & Closed Sets

Open Set: Set E is open iff $\forall x \in E, \exists r > 0,\ s.t.\ (x-r, x+r) \subseteq E$

1. **$\cup$ of open sets**: (E_n) are open sets (Infinite). $\Rightarrow$ $\cup E_n$ is open.
2. **$\cap$ of open sets**: (E_n) are open sets (Finite). $\Rightarrow$ $\cap E_n$ is open.
3. **Find Open Set**:
   If $f: \mathbb{R} \to \mathbb{R}$ continuous $\Rightarrow$ $\{x \in \mathbb{R} : f(x) > \lambda$ (or $< \lambda$) $\}$ is open.
4. **Structure of open sets**: $\Downarrow^{\ominus}$ can be disjoint Hint: $I_k' = \cup I_n \setminus \cup_{i=1}^{k} I_i'$
   E is a nonempty open set $\Leftrightarrow E = \bigcup I_n$, I_n is open interval. (Unique)

Closed Set: Set $F \subseteq \mathbb{R}$ is closed iff $\mathbb{R} \setminus F$ is open.

1. **$\cup$ of closed sets**: (F_n) are closed sets (Finite). $\Rightarrow$ $\cup F_n$ is closed.
2. **$\cap$ of closed sets**: (F_n) are closed sets (Infinite). $\Rightarrow$ $\cap F_n$ is closed.
3. **Find Closed Set**:
   If $f: \mathbb{R} \to \mathbb{R}$ continuous $\Rightarrow$ $\{x \in \mathbb{R} : f(x) \le \lambda$ (or $\ge \lambda$) (or $= \lambda$) $\}$ is closed.
4. **Sequencially Closed**:
   If F is closed set, $\Leftrightarrow$ $\forall (x_n)_{n\in\mathbb{N}} \in F$ convergent. $x = \lim x_n \in F$

$^{\oplus}$ **Open Set $\to \bigcup$ Closed Set**: Nonempty open set $E \to$ union of *disjoint closed* intervals. Hint:[1] Structure of open set. [2] each open interval $(a,b) = \cup[a+\frac{1}{n}, b-\frac{1}{n}]$. [3] closed set→disjoint.

$^{\oplus}$ **Unbounded→ $\cup$ Bounded**: Any closed / open set $F \to$ countable union of *bounded* closed / open sets. (Hint: $K_n = F \cap [-n,n]$ $K_n = F \cap (-n,n)$)

$^{\oplus}$ **Both Open & Closed**: Only $\emptyset$ and $\mathbb{R}$ are both open and closed.

7 Lebesgue Outer Measure || Lebesgue Measurable || Lebesgue Measure

8 Lebesgue Integral

8.1 Step Function || Characteristic Function || Definition of Integral

Characteristic Function: $\chi_E(x) := \{1, x \in E\ ;\ 0, x \notin E\quad ;\quad E \subset \mathbb{R}$

Simple Integral Definitions: $\int_{\mathbb{R}} \chi_I(x)\,dx = \lambda(I)$ || $\int_{\mathbb{R}} c\chi_I(x)\,dx = c\lambda(I)$ || $\int_{\mathbb{R}} \sum c_j \chi_{I_j}(x) dx = \sum c_j \lambda(I_j)$

Step Function: $f: \mathbb{R} \to [0,\infty)$ is a (non-negative) step function if $f(x) = \sum c_j \chi_{I_j}(x)$ for some c_j and χ_{I_j}

· **Property**: If f is a step function $\Rightarrow$ $\exists c_n, I_n$ s.t. I_n *disjoint* and $f(x) = \sum c_n \chi_{I_n}(x)$

9 Appendix

Inequation: *Useful Property*: If $\forall \varepsilon > 0\ a \le b + \varepsilon \Rightarrow a \le b$

Triangular	$\lvert x+y\rvert \le \lvert x\rvert + \lvert y\rvert$	$\lvert x-y\rvert \ge \lvert x\rvert - \lvert y\rvert$	$\lvert\lvert x\rvert - \lvert y\rvert\rvert \le \lvert x \pm y\rvert$.
AM-GM & Cauchy	$a+b \ge 2\sqrt{ab}, a>0, b>0$	$a^2+b^2 \ge 2ab$	$(\sum a_n b_n)^2 \le \sum a_n^2 \sum b_n^2$
Bernoulli's Inequality	$(1+\delta)^\alpha \le 1+\delta\alpha, \alpha \in (0,1], \delta \ge -1$		$(1+\delta)^\alpha \ge 1+\delta\alpha, \alpha \in [1,+\infty), \delta \ge -1$
常见变换技巧 1	$^{\ominus}\lvert sin(x)\rvert \le \lvert x\rvert$ Hint:MVT	$^{\ominus}ln(x) < x, x>0$ Hint: MVT	$^{\ominus}1-cos(x) < \lvert x\rvert$ Hint: MVT
常见变换技巧 2	$^{\ominus}\frac{1}{k} \le ln(x) - ln(x-1)$ Hint:$\int_{k-1}^{k}\frac{dx}{x}$	$\lvert a_n\rvert\lvert x-c\rvert^n = \lvert a_n\rvert\rho^n \cdot \frac{\lvert x-c\rvert^n}{\rho^n}$, $\lvert x-c\rvert < \rho < R$	

$^{\oplus}$ 和差化积/积化和差: s+s=sc,s-s=cs,c+c=cc,c-c=-ss || e.g. $sinA + sinB = 2sin(\frac{A+B}{2})cos(\frac{A+B}{2})$ $sinAsinB = -\frac{1}{2}[cos(A+B) - cos(A-B)]$ Hint: 和差角公式相加减

· $sin(2x) = 2sin(x)cos(x)$ $cos(2x) = 2cos^2(x) - 1 = 1 - sin^2(x) = cos^2(x) - sin^2(x)$ $cos(x) = \sqrt{\frac{1+cos(2x)}{2}}$ $sin(x) = \sqrt{\frac{1-cos(2x)}{2}}$

Hyperbolic Func: $sinh(x) = \frac{e^x - e^{-x}}{2}$ $cosh(x) = \frac{e^x + e^{-x}}{2}$ $cosh^2(x) - sinh^2(x) = 1$ $[sinh(x)]' = cosh(x)$ $[cosh(x)]' = sinh(x)$

· $\sinh(x \pm y) = \sinh(x)\cosh(y) \pm \cosh(x)\sinh(y)$ $\cosh(x \pm y) = \cosh(x)\cosh(y) \pm \sinh(x)\sinh(y)$

Common Int: $\int tan(x)dx = -ln\lvert cos(x)\rvert$; $\int cot(x)dx = ln\lvert sin(x)\rvert$; $\int sec(x)dx = ln\lvert sec(x) + tan(x)\rvert$; $\int csc(x)dx = ln\lvert csc(x) - cot(x)\rvert = ln\lvert tan(\frac{x}{2})\rvert$

Common Der: $[tan(x)]' = sec^2(x)$; $[cot(x)]' = -csc^2(x)$; $[sec(x)]' = sec(x)tan(x)$; $[csc(x)]' = -csc(x)cot(x)$; $[arctan(x)]' = \frac{1}{1+x^2}$.

$1 + tan^2(x) = sec^2(x)$; $1 - csc^2(x) = cot^2(x)$; $csc^2(x) - cot^2(x) = 1$. $[arcsin(x)]' = \frac{1}{\sqrt{1-x^2}}$; $[arccos(x)]' = -\frac{1}{\sqrt{1-x^2}}$

Interchanging the Order of Summation: $\sum_{n=1}^{p}\sum_{m=1}^{q} f(n,m) = \sum_{m=1}^{q}\sum_{n=1}^{p} f(n,m)$ || $^{\oplus}\sum_{k=0}^{n}\sum_{i=0}^{k} = \sum_{i=0}^{k}\sum_{k=i}^{n}$ Hint: 列出来看 or 画图

$^{\oplus}$ **Summation By Parts**: $\sum_{k=1}^{n} a_k(b_{k+1} - b_k) = (a_{n+1}b_{n+1} - a_1 b_1) - \sum_{k=1}^{n}(a_{k+1} - a_k)b_{k+1}$, $\forall (a_k)_{k=1}^{n+1}, (b_k)_{k=1}^{n+1}$

$\Rightarrow \sum_{k=1}^{n} a_k b_k = a_{n+1}\sigma_{n+1} - \sum_{k=1}^{n}(a_{k+1} - a_k)\sigma_{k+1}$, $\forall (a_k)_{k=1}^{n+1}, (b_k)_{k=1}^{n+1}$ $\sigma_k := \sum_{j=1}^{k-1} b_j$

$\mathbf{q^n - 1}$: $q^n - 1 = (q-1)(1 + q + q^2 + \cdots + q^{n-1})$ || $\mathbf{1 \pm x^3}$: $1 \pm x^3 = (1 \pm x)(1 \mp x + x^2)$

Useful Conterexample: $(1/2)^{\frac{1}{n}} \subseteq [1/2, 1)$

$^{\oplus}$ **Common Power Series**: $\sum_{n=0}^{\infty} x^n = (1-x)^{-1}$ $\sum_{n=1}^{\infty} nx^{n-1} = (1-x)^{-2}$ $\sum_{n=2}^{\infty} n(n-1)x^{n-2} = 2(1-x)^{-3}$ $R = 1$ for all.

$^{\oplus}$ $\sum_{n=0}^{\infty}\frac{x^n}{n!} = e^x$ $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{x^n}{n} = ln(x+1)$ $\sum_{n=1}^{\infty}\frac{x^n}{n} = -ln(1-x)$ $R = \infty$ for all.

Sum of $\mathbf{n}, \mathbf{n^2}, \mathbf{n^3}$: $\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$ $\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$ $\sum_{k=1}^{n} k^3 = [\frac{n(n+1)}{2}]^2$

Countably Approximation Tools: $\sum_{n=1}^{\infty} 2^{-n}\varepsilon = \varepsilon$

De Morgan's Laws: $(A \cap B)^C = A^C \cup B^C$ $(A \cup B)^C = A^C \cap B^C$

$^{\oplus}$ $E \setminus F = F^C \cap E$ is open (i.e. open - closed = open) || $F \setminus E = E^C \cap F$ is open (i.e. closed - open = closed)

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10 Integration Theory & Lebesgue Integration

10.1 Lebesgue Outer Measures

Countably Additivity: $(E_n)_{n\in\mathbb{N}}$ disjoint sets, m be measure. $\Rightarrow$ Countably Additive if: $m(\cup_n E_n) = \sum m(E_n)$

Lebesgue Outer Measure: $m^*(E) := \inf\{\sum_{n=1}^{\infty}\lambda(I_n) \mid E \subseteq \cup_{n=1}^{\infty} I_n, I_n\ is\ a\ open\ interval\}$ ps: 本质是寻找到最小的区间可以覆盖 E, 用前面 λ 的定义来定义" 长度"

· **Remark**: If it's impossible to find $(I_n)_{n\in\mathbb{N}}$ s.t. $\sum \lambda(I_n) < \infty \Rightarrow m^*(E) = \infty$ || Can interpret m^* as $m^*: \mathcal{P}(\mathbb{R}) \to [0,\infty)$

1. **Lebesgue Outer Measure (Open)**: $m^*(E) = \inf\{\lambda(U), E \subseteq U, U\ is\ open\ set\}$ ps:$\lambda(U) = \sum \lambda(I_n)$
2. **Lebesgue Outer Measure (Interval)**: If $J \subseteq \mathbb{R}$ is interval. $m^*(J) = \lambda(J)$

3. **Useful Tool|By def**: $\forall E, \exists$ open interval $(I_n)_{n\in\mathbb{N}}$ s.t. $E \subseteq \cup I_n$ and $\sum \lambda(I_n) < m^*(E) + \varepsilon$
4. **Useful Tool|By def**: $\forall E, \exists$ open set U s.t. $E \subseteq U$ and $m^*(U) < m^*(E) + \varepsilon$
5. **Useful Tool|Plus|By def**: $\forall E, \exists$ open intervals $(I_n)_{n\in\mathbb{N}}$, satisfy [1] 第 3 条 [2] $I_n \cap E \neq \emptyset$ [3] $\lambda(I_n) < \delta$ [4]
6. **Futher|$m^{**}(E)$**: $m^*(E) = m^{**}(E) := \inf\{\sum \lambda(I_n) \mid E \subseteq \cup I_n, I_n\ interval\}$ || $m^*(E) \leq \sum \lambda(I_n)$, if $E = \cup I_n$

Properties of Lebesgue Outer Measure: Let Set $E, F\ s.t.\ E \subseteq F \subset \mathbb{R}$. Emptyset: $\emptyset$

1. **Null Empty Set**: $m^*(\emptyset) = 0$.
2. **Monotonicity**: $m^*(E) \leq m^*(F)$.
3. **Countable Subadditivity**: $m^*(\cup_{n=1}^{\infty} E_n) \leq \sum_{n=1}^{\infty} m^*(E_n)$.
4. Countable set $E \iff m^*(E) = 0$ Hint: $E = \cup_{j=1}^{\infty}\{x_j\}$ || $\exists m^*(E) = 0, E$ uncountable (e.g.Cantor Set)

10.2 Countable Additive for Lebesgue Outer Measure

Def of $dist(E,F)$: $dist(E,F) := \inf\{|x-y| : x \in E, y \in F\}$

1. **Basic**: If $dist(E,F) > 0 \Rightarrow m^*(E \cup F) = m^*(E) + m^*(F)$.
   · [1] E, F are nonempty *closed* sets; [2] $E \cap F = \emptyset$; [3] $E\ bounded$. $\Rightarrow dist(E,F) > 0$
2. **Closed|Open|Interval Sets**: If $(K_n)_{n\in\mathbb{N}}$ is *disjoint*, (Closed|Open|Interval), *bounded* sets $\Rightarrow m^*(\cup_n^{\infty} K_n) = \sum_{n=1}^{\infty} m^*(K_n)$

10.3 Lebesgue Measurable Sets

Lebesgue Measurable:

1. **Version 1**: A set $E \subseteq \mathbb{R}$ is *Lebesgue measurable* if $\forall \varepsilon > 0, \exists$ open set $U \subseteq \mathbb{R}$ s.t. $E \subseteq U$ and $m^*(U \setminus E) < \varepsilon$ ps: 被 open set 从外" 恰好" 包住
2. **Version 2**: A set $E \subseteq \mathbb{R}$ is *Lebesgue measurable* if $\forall \varepsilon > 0, \exists$ closed set $U \subseteq \mathbb{R}$ s.t. $U \subseteq E$ and $m^*(E \setminus U) < \varepsilon$ ps: 被 closed set 从内" 恰好" 包住

· $\mathcal{M}$:= collection of all Lebesgue measurable sets of $\mathbb{R}$.

Property of Lebesgue Measurable Sets:

1. **Closed Set|Open Set|Interval**: Any closed set / open set / interval is Lebesgue measurable.
2. **Union $\cup$|Intersection $\cap$**: if $(E_n)_{n\in\mathbb{N}}$ is *Lebesgue Measurable* $\Rightarrow$ [1] $E := \cup_{n=1}^{\infty} E_n$ [2] $E := \cap_{n=1}^{\infty} E_n$ is *Lebesgue Measurable*.
3. **Complement**: if $E \subseteq \mathbb{R}$ is *Lebesgue Measurable* $\Rightarrow \mathbb{R} \setminus E$ is also *Lebesgue Measurable*.
4. **Measure Zero Set**: $m^*(E) = 0 \iff E$ Lebesgue measurable.
5. **Translation Invariance**: $m^*(E) = m^*(x+E)$ || If E is Lebesgue Measurable $\Rightarrow x + E$ also.

10.4 Countable Additive & Monotone Convergence for Lebesgue Measure

Lebesgue Measure: *Lebesgue Measure* is a function $m : \mathcal{M} \to [0,\infty)$ s.t. $m(E) := m^*(E), \forall m \in \mathcal{M}$ ps: $\mathcal{M}$ 是所有 Lebesgue 可测集合的集合

· **Property**: ⋆ *All properties of Lebesgue Outer Measure hold for Lebesgue Measure.* ⋆ ps: 原来是 $m^* : \mathcal{P}(\mathbb{R}) \to [0,\infty)$, 现在是 $m : \mathcal{M} \to [0,\infty)$

Countable Additivity for Measure: If $(E_n)_{n\in\mathbb{N}}$ *disjoint* & *Lebesuge Measurable*. $\Rightarrow m(\cup_{n=1}^{\infty} E_n) = \sum_{n=1}^{\infty} m(E_n)$

Monotone Convergence for Measure: If $(E_n)_{n\in\mathbb{N}}$ *Lebesuge Measurable*.

1. **Increasing↑**: If $E_n \subseteq E_{n+1} \Rightarrow m(\cup_{n=1}^{\infty} E_n) = \lim_{n\to\infty} m(E_n)$
2. **decreasing↓**: If $E_{n+1} \subseteq E_n, m(E_1) < \infty \Rightarrow m(\cap_{n=1}^{\infty} E_n) = \lim_{n\to\infty} m(E_n)$ (Hint: $F_n := E_n \setminus E, E = \cap E_n$)

Vitali Set: $\exists A \subseteq \mathbb{R}$ and $\exists$ disjoint $E_j \subseteq \mathbb{R}$ s.t. A is not *Lebesgue measurable* and $m^*(\cup E_j) \neq \sum_j m^*(E_j)$ || ps: Vitali Set is a set that is not Lebesgue measurable.

· Example: